

# Language Identification using Transformer-based Architectures: A Comparative Study with mBERT and downstream classifiers

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## Abstract

Language Identification (LID) is a foundational task in Natural Language Processing (NLP) that determines the language of a given text. In this study, we evaluate three Transformer-based approaches to LID using the WiLI-2018 benchmark dataset. We employ mBERT as a shared encoder and compare the effectiveness of (i) a linear classification head as baseline, (ii) an MLP classifier, and (iii) a CNN classifier. Our results suggest that more complex downstream architecture choices do not yield notable performance improvements in LID tasks. We evaluate all three models on standard metrics, provide an in-depth error analysis, and express actionable future work directions.

## 1 Introduction

Language identification (LID) is the task of determining the language in which a given piece of text is written, and it plays a critical role in multilingual Natural Language Processing (NLP) systems such as machine translation, and information retrieval. Traditional approaches relied on character  $n$ -grams (Hughes et al., 2006), but recent advances in deep learning, especially Transformer-based models, have greatly improved performance.

Correct LID is critical not only to improve NLP performance, but also to ensure linguistic equity. When language models fail to distinguish smaller languages from closely related high-resource ones (e.g., Malay vs. Indonesian), they risk reinforcing digital inequities and reducing the visibility of underrepresented language communities (Nomoto, 2023).

This work explores how different downstream classifiers perform when paired with a shared multilingual encoder (mBERT). We assess three configurations: i) mBERT + Linear classifier, ii) mBERT + MLP, iii) mBERT + CNN. Our goal is to understand whether classifier complexity improves lan-

guage discrimination ability, especially on a large multilingual dataset like WiLI.

## 2 Related Work

Early surveys of written LID outlined the strengths and weaknesses of statistical and rule-based approaches. Hughes et al. (2006) reviewed decades of methods, from character  $n$ -grams to kernel-based classifiers, and identified persistent challenges such as limited support for minority languages, lack of standardized corpora, and reliance on large labeled resources. These limitations framed LID as a mature yet incomplete task, with many of these challenges remain relevant today.

With the rise of neural networks, researchers began exploring architectures capable of learning language-discriminative features directly from text. Tian et al. (2002); Tian and Suontausta (2003) applied multilayer perceptrons (MLPs), while Jacek et al. (2016b,a) showed that convolutional neural networks (CNNs) effectively capture character-level information. Li et al. (2018) further introduced adversarial learning for improved domain robustness, though these models still depended on large supervised datasets.

The main dataset for this project, WiLI-2018 dataset (Thoma, 2018) provides a standardized multilingual benchmark (235 languages) that led to comparative evaluation of neural LID systems. More recently, multilingual Transformers such as mBERT (Devlin et al., 2018) and XLM-R have transformed the task by enabling strong zero-shot transfer. Pires et al. (2019) showed that mBERT performs competitively on unseen languages without fine-tuning.

Beyond overall accuracy, recent studies have examined embedding behavior. Wyawahare (2023) compared MLP, CNN, and LSTM classifiers using FastText and Sentence Transformer embeddings, visualizing clustering with t-SNE and showing how

embedding dimensionality influences accuracy.

Unlike prior work that fine-tunes the encoder, we begin by assessing mBERT’s zero-shot LID capacity by attaching a randomly initialized linear classification head. Subsequently, we explore the impact of classifier complexity by finetuning Linear, MLP, and CNN heads on top of a mBERT encoder, isolating their impact from encoder adaptation and providing a complementary perspective on Transformer-based LID.

### 3 Dataset

This section provides an overview of the two datasets used in our empirical testing.

#### 3.1 In Domain: WiLI

The WiLI-2018 dataset (Thoma, 2018) contains 1,000 Wikipedia paragraphs for each of 235 languages. Since mBERT supports 104 languages<sup>1</sup>, we constructed a subset covering 102 overlapping languages by matching WiLI language codes to mBERT’s supported *Wikipedias*.

Following Thoma (2018), we note that some entries exhibit label noise due to copied text, mis-translations, or mixed-language content. Despite this, WiLI remains a widely used benchmark for multilingual LID.

We further analyzed our subset for character diversity and average paragraph length, confirming patterns consistent with the original WiLI release (see Appendix A).

#### 3.2 Out of Domain: Language Identification

The Language Identification dataset was utilized as out of domain dataset (Papaluca, 2021), which contains 10,000 sentences from 20 different languages. The dataset is a combination of Multilingual Amazon Reviews Corpus (Keung et al., 2020), XNLI (Conneau et al., 2018), and data labeled as STSb Multi MT. This dataset was used to ensure robustness, i.e. to investigate whether the models’ LID capabilities generalize beyond the domain of Wikipedia articles they were trained on.

## 4 Methodology

This section provides an overview of the methodology of the experiments we conducted. For details about the technical configuration, see Appendix C

<sup>1</sup><https://github.com/google-research/bert/blob/master/multilingual.md#list-of-languages>

### 4.1 Model

The model we use for this project is multilingual BERT (mBERT) (Devlin et al., 2018), a multilingual encoder based model trained using masked language modeling (MLM<sup>2</sup>) and next-token prediction tasks across more than 100 languages. It facilitates using languages with non-Latin alphabets, and additionally supports Thai and Mongolian, bringing the total amount of languages supported to 104 (see Appendix B for details about mBERT and associated languages).

### 4.2 Dataset

As discussed in Section 3, our work considers the subset of 102 languages of the WiLI-2018 dataset, with a total of 102,000 paragraphs of text. We split the data into training (50% → 51,000 instances), validation (25% → 25,500 instances), and test (25% → 25,500 instances) sets, with each language having equal representation in each set (i.e. stratified by language).

### 4.3 Encoding

The standard BertTokenizer is used to tokenize the input sequence, and pads/truncates it to 512 tokens. To obtain the encoding for each input sentence, the final hidden state of the mBERT model given the [CLS] token is used. This 768-dimensional embedding vector is explicitly trained to capture information about the entire sequence.

### 4.4 Classification Head

To classify the 768-dimensional embedding vector corresponding to each input text, we extend mBERT with one of three different classification heads:

- **Linear:** Single linear layer using HuggingFace’s BertForSequenceClassification
- **MLP:** Two-layer MLP (custom PyTorch implementation)
- **CNN:** Two-layer one-dimensional CNN, using max-pooling, and a fully connected output layer (custom PyTorch implementation)

### 4.5 Training

During training, for all three model configurations, both the mBERT model and classification head were trained. Since mBERT is a pre-trained model

<sup>2</sup>Known as *Cloze* task in existing literature (Taylor, 1953)

and the classification head were randomly initialized, two different learning rates were used. To ensure reproducibility, each model configuration was run with random seeds 32, 42, and 52. More technical details are discussed in Appendix C.4.

## 5 Results

This section provides the results of our empirical testing. Note that accuracy and macro-averaged recall are identical since the dataset is perfectly balanced, making metrics reflect the same proportion. All empirically obtained metrics are shown in Appendix D.

### 5.1 In-Domain Data

Table 1 shows the performance across all setups. It can be seen that the performance for all metrics across fine-tuned models is consistently high, with a small standard deviation between seeds.

The Linear classification head slightly outperforms the other models in average performance metrics. However, for the fine-tuned models none of the four metrics a statistical significant ( $\alpha = 0.05$ ,  $t$ -test) difference exists between any pair of models. The unfine-tuned mBERT performs significantly worse than it’s fine-tuned counterparts.

| Model         | Acc/Rec           | Prec              | F1                |
|---------------|-------------------|-------------------|-------------------|
| mBERT+Linear* | 0.011 $\pm$ 0.000 | 0.004 $\pm$ 0.002 | 0.003 $\pm$ 0.001 |
| mBERT+Linear  | 0.969 $\pm$ 0.004 | 0.973 $\pm$ 0.002 | 0.969 $\pm$ 0.004 |
| mBERT+MLP     | 0.964 $\pm$ 0.003 | 0.969 $\pm$ 0.002 | 0.962 $\pm$ 0.005 |
| mBERT+CNN     | 0.966 $\pm$ 0.003 | 0.969 $\pm$ 0.004 | 0.966 $\pm$ 0.004 |

Table 1: Performance of classification heads on the WiLI subset. Average over the three seeds, and standard deviation are shown. Precision, recall, and F1 use macro-averaging. \*zero-shot: without fine-tuning

### 5.2 Out of Domain Data

Table 2 shows the performance across all setups on a different domain than the model was trained on (Section 3.2). The performance for all metrics across fine-tuned models is still consistently high, yet slightly lower and with more standard deviation between seeds than when evaluating on similar data the model was trained on.

Both the Linear and MLP outperformed the CNN classification head by a larger margin than on the in-domain data. However, just as before, for the fine-tuned models none of the four metrics a statistical significant ( $\alpha = 0.05$ ,  $t$ -test) difference exists between any pair of models. Only accuracy/recall shows a statistical significant ( $\alpha = 0.05$ ,  $t$ -test)

difference for each classification head compared to their in-domain test set results.

| Model        | Acc/Rec           | Prec              | F1                |
|--------------|-------------------|-------------------|-------------------|
| mBERT+Linear | 0.945 $\pm$ 0.012 | 0.997 $\pm$ 0.001 | 0.968 $\pm$ 0.007 |
| mBERT+MLP    | 0.950 $\pm$ 0.021 | 0.992 $\pm$ 0.001 | 0.969 $\pm$ 0.013 |
| mBERT+CNN    | 0.906 $\pm$ 0.030 | 0.994 $\pm$ 0.002 | 0.940 $\pm$ 0.021 |

Table 2: Performance of classification heads on the Language Identification subset. Average over the three seeds, and standard deviation are shown. Precision, recall, and F1 use macro-averaging.

## 6 Analysis

This section analyzes the results we obtained.

### 6.1 Model Performance

In-domain findings only showed a small, statistically insignificant, performance difference between the three model configurations. This suggests that for LID using mBERT as an encoder, more complex classifiers do not present a benefit over simpler classification models.

Out of domain results support these findings. While accuracy drops slightly for all models, the linear head remains competitive with the others. Notably, the CNN head experiences the highest reduction in accuracy of 0.060, compared to 0.024 of the linear head, indicating that the CNN is less robust to domain shifts.

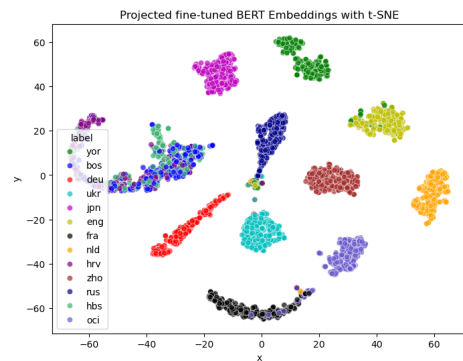


Figure 1: t-SNE projections of the 768-dimensional mBERT embeddings of all texts present in the WiLI test set for a language, after fine-tuning mBERT (on mBERT + Linear).

When observing t-SNE projections of the 768-dimensional mBERT embeddings of texts in Figure 1, we observe the explanation for this behavior. mBERT embeddings already capture language specific linguistic properties rather well, creating clear clusters in the embedding space after fine-tuning

(see Appendix E). This easy differentiability explains the lack of classification performance improvement when moving from a linear classifier to more complex classifiers.

While performance remains high on the novel Language Identification subset, the statistical significant performance degradation compared to the WiLI test set points to overfitting. Mitigating this might be considered in future work.

## 6.2 Error Analysis

Overall, the models perform well on most language pairs, but certain systematic misclassifications reveal underlying linguistic or model-specific weaknesses. Confusion matrices for all model configurations are found in Appendix F.

### 6.2.1 General Trends in Misclassifications

Figure 2 shows a closer look into the matrix generated for the Linear classifier. This confusion matrix shows a trend present in all model configurations (refer to Appendix F): a strong diagonal dominance, indicating high accuracy and limited off-diagonal confusion. This confirms that, regardless of head architecture, mBERT maintains robust cross-lingual separation for most languages.

Systematic errors primarily occur among closely related South Slavic varieties: Bosnian, Croatian, and Serbo-Croatian. These form a pluricentric language with near-identical grammar and morphology (Bailyn, 2010; Bunčić, 2021).

Less frequent confusions involve linguistically or geographically related pairs such as German–Bavarian (Rowley, 2011) and Malay–Indonesian (Sneddon, 2004), which likely reflect genuine linguistic overlap rather than model failure.

A few Yoruba samples mislabeled as English likely correspond to known dataset noise (Appendix A).

### 6.2.2 Model-Specific Observations

All classifiers show similar South Slavic confusion patterns but differ in severity. The MLP (Figure 6a) exhibits the highest overlap, frequently misclassifying Croatian and Serbo-Croatian as Bosnian and producing the broadest spread of minor errors (e.g., Indonesian–Malay, Bavarian–German). The CNN (Figure 6b) shows milder confusion within the same cluster but greater overall robustness, reflected by stronger diagonal dominance. The Linear model (Figure 7) performs comparably, with

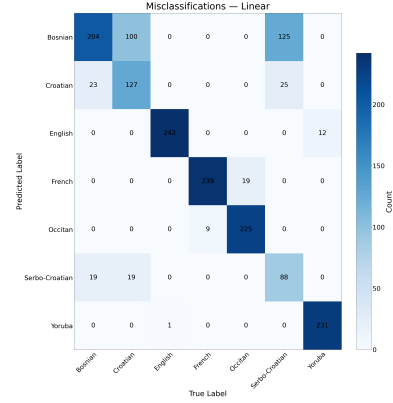


Figure 2: Zoomed-in confusion matrix for the mBERT + Linear classifier.

136 total misclassifications, effectively separating distant languages and even some similar pairs (e.g., Malay–Indonesian). These results suggest that classifier complexity offers minimal benefit once features are extracted by mBERT.

## 6.3 Actionable Future Directions

Future work could use language models trained on combined data from such related varieties, for example: *BERTiC* (Ljubešić and Lauc, 2021) was pre-trained on over eight billion tokens of Bosnian, Croatian, Montenegrin, and Serbian text, and demonstrated that unified training can effectively capture shared structure within the HBS macro-language. Similar challenges have been reported for other languages, including Bavarian and German dialects (Peng et al., 2024; Blaschke et al., 2025), Indonesian–Malaysian (Abdiansah and Rizqie, 2023), and regional French varieties (Maillard and Khan, 2024). Our results, combined with the existing works discussed above, indicate that misclassifications between closely related languages reflect real linguistic overlap rather than model failure. This highlights the importance of developing multilingual or regionally sensitive models that can capture gradual differences across dialect boundaries.

## 7 Conclusion

We presented a comparative study on Transformer-based LID, using mBERT as a shared encoder and testing three classification paradigms. Our results suggest that incorporating more complex classification heads did not yield notable performance improvements in LID tasks. Future work could explore using larger multilingual models, or addressing class imbalance in low-resource languages.



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## A WiLI Analysis

This appendix provides the detailed procedure for constructing the WiLI-mBERT subset and supplementary dataset analyses.

**Subset construction** We use the WiLI-2018 dataset (Thoma, 2018), which contains 1,000 Wikipedia paragraphs per language, spanning 235 languages. Since mBERT only supports 104 languages<sup>3</sup>, we matched the mBERT languages to their respective Wiki codes using the list of languages of Wikipedias that have been created as separate subdomains of wikipedia.org, ordered by number of articles. Then these Wiki codes were compared to those available in the WiLI-2018 dataset. From this we were able to create our working subset of WiLI which contains 102 languages (we were unable to find a Wiki code for Traditional Chinese, and WiLI does not contain Piedmontese)

**Data characteristics** Each sample in WiLI is a Wikipedia paragraph of text containing at least 140 Unicode characters, providing diverse linguistic structure across samples.

Each language can contain words or sentences from another language (direct quotes) or special characters (like those used to help with pronunciation), since they are excerpts from Wikipedia. However, there can be errors or difficulties in the labelling. For example, Thoma (2018) discusses: misclassified sources (especially for literature), instances of just copy paste across Wikipedias without translations, mistranslations, quotes in foreign languages, and writing-style/topic biases.

One such example could be the following paragraph which is labelled as Yoruba, but is vastly written in English.

*"Klára Zakopalová (Àdàk:IPA-cs; born as Klára Koukalová on 24 February 1982 in Prague, Czechoslovakia) is a Czech professional female tennis player. She was born and still lives in Prague. je agba tennis."*(instance 31924 of the train data).

**Subset analysis** Following Thoma (2018), we analyzed our WiLI-mBERT subset further in terms of character diversity and paragraph length. Figure 3 shows the number of unique characters required to cover 99% of all occurrences per language ( $|C_{0.99}|$ ). Figure 4 highlights the variation in average paragraph length across languages. The distributions

confirm that our subset preserves the variation observed in the original WiLI dataset.

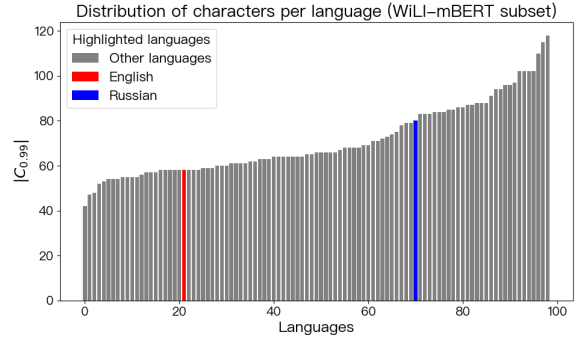


Figure 3: Distribution of characters per language ( $|C_{0.99}|$ ) for our WiLI-mBERT subset. English is highlighted in red with  $|C_{0.99}| = 58$ , Russian in blue with  $|C_{0.99}| = 80$ . Languages with very large character sets (e.g., Chinese (2814 characters), Japanese (1630 characters), Korean (1158 characters)) are excluded for readability.

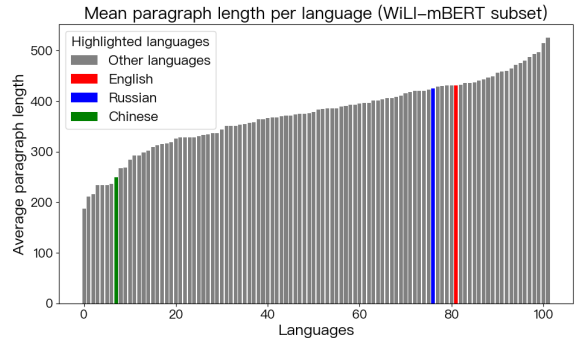


Figure 4: Mean paragraph length per language in our WiLI-mBERT subset. English is highlighted in red, Russian in blue, and Chinese in green. One can see that the average length of a paragraph is vastly different between languages (Smallest average length: 187.16 Minangkabau vs. Largest average length: 525.88 Uzbek); however, most appear to be in the 300-400 range, which corresponds with our finding of 375 characters being the the mean paragraph length averaged over all languages.

<sup>3</sup><https://github.com/google-research/bert/blob/master/multilingual.md#list-of-languages>

## B mBERT languages

| mBERT Language       | Wiki Code | Current Articles |
|----------------------|-----------|------------------|
| Afrikaans            | af        | 126,374          |
| Albanian             | sq        | 104,246          |
| Arabic               | ar        | 1,280,180        |
| Aragonese            | an        | 50,024           |
| Armenian             | hy        | 322,417          |
| Asturian             | ast       | 137,579          |
| Azerbaijani          | az        | 207,460          |
| Bashkir              | ba        | 63,905           |
| Basque               | eu        | 471,995          |
| Bavarian             | bar       | 27,199           |
| Belarusian           | be        | 256,058          |
| Bengali              | bn        | 175,812          |
| Bishnupriya Manipuri | bpy       | 25,092           |
| Bosnian              | bs        | 95,793           |
| Breton               | br        | 89,288           |
| Bulgarian            | bg        | 305,738          |
| Burmese              | my        | 109,598          |
| Catalan              | ca        | 781,093          |
| Cebuano              | ceb       | 6,116,073        |
| Chechen              | ce        | 602,390          |
| Chinese (Simplified) | zh        | 1,500,405        |
| Chuvash              | cv        | 57,886           |
| Croatian             | hr        | 227,686          |
| Czech                | cs        | 577,081          |
| Danish               | da        | 310,655          |
| Dutch                | nl        | 2,197,125        |
| English              | en        | 7,057,584        |
| Estonian             | et        | 254,787          |
| Finnish              | fi        | 603,378          |
| French               | fr        | 2,709,694        |
| Galician             | gl        | 226,677          |
| Georgian             | ka        | 185,233          |
| German               | de        | 3,051,575        |
| Greek                | el        | 258,272          |
| Gujarati             | gu        | 30,666           |
| Haitian              | ht        | 71,053           |
| Hebrew               | he        | 382,681          |
| Hindi                | hi        | 166,599          |
| Hungarian            | hu        | 560,809          |
| Icelandic            | is        | 60,550           |
| Ido                  | io        | 59,405           |
| Indonesian           | id        | 743,356          |
| Irish                | ga        | 62,347           |
| Italian              | it        | 1,935,325        |
| Japanese             | ja        | 1,472,968        |
| Javanese             | jv        | 74,644           |
| Kannada              | kn        | 33,957           |
| Kazakh               | kk        | 240,132          |
| Kirghiz              | ky        | 76,020           |
| Korean               | ko        | 721,336          |
| Latin                | la        | 140,511          |
| Latvian              | lv        | 137,480          |
| Lithuanian           | lt        | 223,609          |
| Lombard              | lmo       | 78,810           |
| Low Saxon            | nds-nl    | 8,038            |
| Luxembourgish        | lb        | 65,706           |
| Macedonian           | mk        | 155,009          |
| Malagasy             | mg        | 100,991          |
| Malay                | ms        | 432,429          |
| Malayalam            | ml        | 87,249           |
| Marathi              | mr        | 100,449          |
| Minangkabau          | min       | 228,514          |
| Mongolian            | mn        | 26,221           |
| Nepali               | ne        | 29,373           |

| mBERT Language      | Wiki Code | Articles  |
|---------------------|-----------|-----------|
| Newar               | new       | 72,586    |
| Norwegian (Bokmal)  | no        | 656,245   |
| Norwegian (Nynorsk) | nn        | 175,541   |
| Occitan             | oc        | 90,169    |
| Persian (Farsi)     | fa        | 1,054,894 |
| Piedmontese         | pms       | 70,523    |
| Polish              | pl        | 1,668,891 |
| Portuguese          | pt        | 1,155,630 |
| Punjabi             | pa        | 59,037    |
| Romanian            | ro        | 516,568   |
| Russian             | ru        | 2,063,491 |
| Scots               | sco       | 34,293    |
| Serbian             | sr        | 710,795   |
| Serbo-Croatian      | sh        | 461,057   |
| Sicilian            | scn       | 26,236    |
| Slovak              | sk        | 255,687   |
| Slovenian           | sl        | 195,049   |
| South Azerbaijani   | azb       | 244,279   |
| Spanish             | es        | 2,061,364 |
| Sundanese           | su        | 62,109    |
| Swahili             | sw        | 101,507   |
| Swedish             | sv        | 2,616,265 |
| Tagalog             | tl        | 48,853    |
| Tajik               | tg        | 115,868   |
| Tamil               | ta        | 176,748   |
| Tatar               | tt        | 503,156   |
| Telugu              | te        | 115,755   |
| Thai                | th        | 176,219   |
| Turkish             | tr        | 646,685   |
| Ukrainian           | uk        | 1,391,010 |
| Urdu                | ur        | 232,406   |
| Uzbek               | uz        | 303,532   |
| Vietnamese          | vi        | 1,296,369 |
| Volapük             | vo        | 44,809    |
| Waray-Waray         | war       | 1,266,766 |
| Welsh               | cy        | 282,455   |
| West Frisian        | fy        | 57,977    |
| Western Punjabi     | pnb       | 74,607    |
| Yoruba              | yo        | 35,792    |

Table 3: mBERT languages with corresponding Wikipedia codes and number of articles. \* = mBERT github page does not specify which Chinese language it takes for Traditional Chinese, so there is no associated wikipedia code or number of articles

The multilingual BERT (mBERT) (Devlin et al., 2018) model was trained using a masked language modeling (MLM) (Taylor, 1953) objective on what used to be the top 100 Wikipedias (by size) in 2018, using full Wikipedia dumps with user and talk pages excluded. Since the number of Wikipedia articles varies drastically between languages (as can be seen in the Articles column of Table 3), data sampling was smoothed to avoid over-representing high-resource languages and under-representing low-resource ones. Specifically, each language’s probability is exponentiated by  $S = 0.7$  before normalizing it again. This reduces the dominance of large Wikipedias (e.g., English) while upweighting smaller ones (e.g., Tagalog).



A single 110k shared WordPiece (Wu et al., 2016) vocabulary is used, with the same weighting scheme applied to token counts. For CJK characters (Chinese, Japanese Kanji, Korean Hanja), spaces are inserted around each character, making them effectively character-tokenized, while other scripts follow a standard pipeline (lowercasing, accent removal, punctuation splitting, whitespace tokenization). The explicit language markers are not included, enabling zero-shot transfer.

The *Multilingual Cased* model we utilize is a later release which supports the aforementioned 100 largest Wikipedias at training time, in addition to Thai and Mongolian. We use this model to ensure we can use languages with non-Latin alphabets, and because it fixes normalization issues in many languages. Table 3 lists all supported languages, their Wikipedia codes, and article counts.

## C Implementation details

This appendix provides a detailed breakdown of the classifier architectures that extend the mBERT encoder. In all configurations, we use the 768-dimensional [CLS] token embedding from the final hidden layer of mBERT as input to the classification head.

The classification heads consist of the following architecture in given order:

### C.1 mBERT + Linear

1. **Input:** 768-dimensional [CLS] token embedding from mBERT
2. **Layer 1:** Linear layer
  - Input: 768-dimensional
  - Output: 102-dimensional
3. **Activation:** Softmax (inside CrossEntropy-Loss)

### C.2 mBERT + MLP

1. **Input:** 768-dimensional [CLS] token embedding from mBERT
2. **Dropout**
3. **Layer 1:** Linear layer
  - Input: 768-dimensional
  - Output: 1,152-dimensional
4. **Activation:** GELU
5. **Layer 2:** Linear layer

- Input: 1,152-dimensional
- Output: 768-dimensional

### 6. Activation: GELU

### 7. Layer 3: Linear layer

- Input: 768-dimensional
- Output: 102-dimensional

### 8. Activation: Softmax (inside CrossEntropy-Loss)

## C.3 mBERT + CNN

### 1. Input: 768-dimensional [CLS] token embedding from mBERT

### 2. Dropout

### 3. Layer 1: 1-dimensional convolutional layer

- Input: 1 channel, 768-dimensional
- Output: 4 channels, 766-dimensional
- Kernel size: 3
- Padding: 0

### 4. Activation: ReLU

### 5. Layer 2: 1-dimensional convolutional layer

- Input: 4 channels, 766-dimensional
- Output: 4 channels, 764-dimensional
- Kernel size: 3
- Padding: 0

### 6. Activation: ReLU

### 7. Pooling: Max pooling

- Input: 4 channels, 764-dimensional
- Output: 4 channels, 382-dimensional
- Window size: 2
- Stride: 2

### 8. Flatten

### 9. Layer 3: Linear layer

- Input: 1,528-dimensional
- Output: 102-dimensional

### 10. Activation: Softmax (inside CrossEntropy-Loss)

#### C.4 Training Hyperparameters (Shared)

During training, both the mBERT model and classification head were trained. Since mBERT is already pre-trained, it requires more gentle fine-tuning than the randomly initialized classification head. Therefore, two different learning rates were used. Since mBERT + Linear makes use of the predefined BertForSequenceClassification model with known good performance using a single learning rate, we only considered the mBERT learning rate for this model.

- Optimizer: AdamW
- Learning rate:
  - mBERT: 5e-5
  - Classifier head: 1e-4
- Weight decay: 0.01
- Batch size: 16
- Epochs: 10 (with early stopping based on evaluation loss metric)

#### C.5 Tokenizer and Input Preprocessing

- Tokenizer: bert-base-multilingual-cased
- Max sequence length: 512
- Truncation and padding applied

#### D All empirical results

| Dataset  | Model  | Seed | Acc/Rec | Prec   | F1     |
|----------|--------|------|---------|--------|--------|
| WiLI     | MLP    | 32   | 0.9629  | 0.9704 | 0.9620 |
|          | MLP    | 42   | 0.9601  | 0.9670 | 0.9563 |
|          | MLP    | 52   | 0.9683  | 0.9703 | 0.9688 |
|          | Linear | 32   | 0.9744  | 0.9755 | 0.9746 |
|          | Linear | 42   | 0.9658  | 0.9700 | 0.9643 |
|          | Linear | 52   | 0.9680  | 0.9722 | 0.9682 |
|          | CNN    | 32   | 0.9656  | 0.9705 | 0.9663 |
|          | CNN    | 42   | 0.9708  | 0.9727 | 0.9713 |
|          | CNN    | 52   | 0.9623  | 0.9632 | 0.9606 |
| Papaluca | MLP    | 32   | 0.9611  | 0.9905 | 0.9752 |
|          | MLP    | 42   | 0.9679  | 0.9937 | 0.9803 |
|          | MLP    | 52   | 0.9204  | 0.9931 | 0.9508 |
|          | Linear | 32   | 0.9277  | 0.9974 | 0.9588 |
|          | Linear | 42   | 0.9515  | 0.9955 | 0.9717 |
|          | Linear | 52   | 0.9555  | 0.9979 | 0.9749 |
|          | CNN    | 32   | 0.8650  | 0.9974 | 0.9119 |
|          | CNN    | 42   | 0.9171  | 0.9937 | 0.9470 |
|          | CNN    | 52   | 0.9360  | 0.9919 | 0.9613 |

Table 4: Performance metrics of all empirical runs on both datasets.

Table 4 shows the performance metrics of all empirical runs. Out of these results Table 1 and Table 2 were calculated.

#### E t-SNE embeddings

In the main body of the paper, Figure 1 showed the 768-dimensional mBERT embeddings after fine-tuning the mBERT + Linear model. To investigate the out-of-the-box differentiation performance of mBERT between languages, we plotted the embeddings of the same texts before fine-tuning. Figure 5 shows how the embeddings of different languages are scattered around in embedding space (possibly words across languages close in meaning close together, however, we did not verify this hypothesis). This is in contrast to Figure 1, which showed clear clusters for different languages. We conclude that fine-tuning the mBERT model for language classification greatly improves differentiability between languages in embedding space.

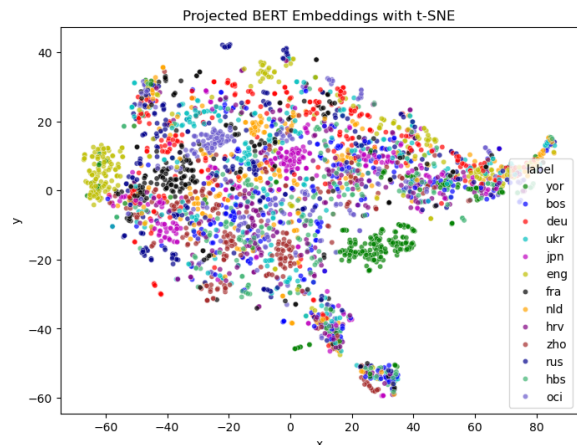
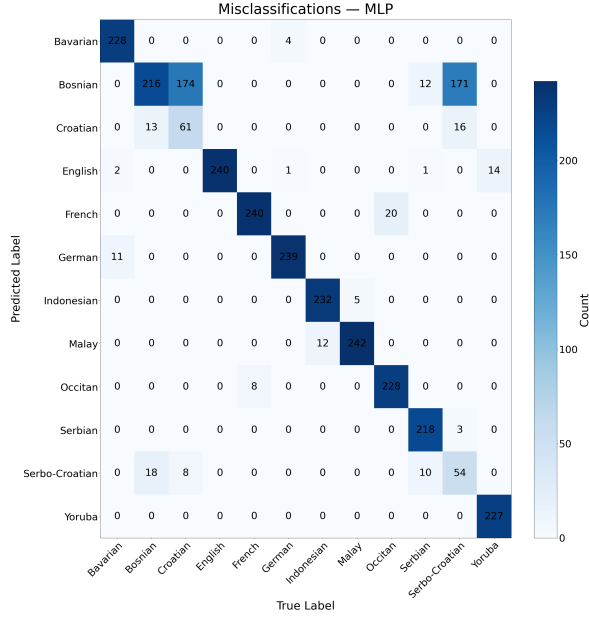


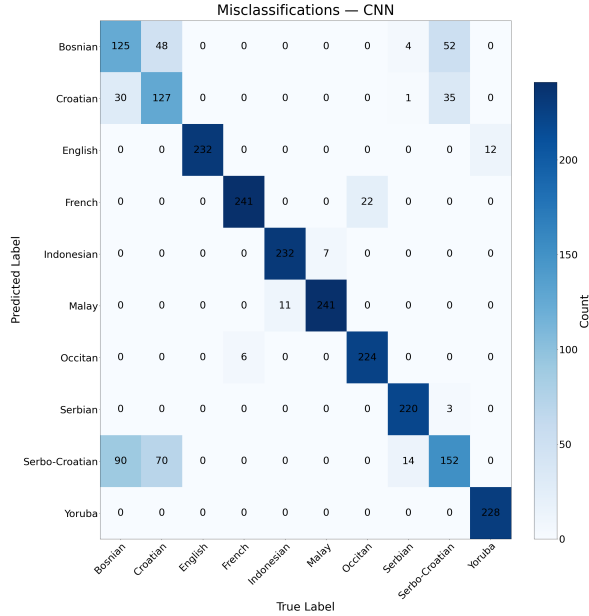
Figure 5: t-SNE projections of the 768-dimensional mBERT embeddings of all texts present in the WiLI test set for a language, before fine-tuning mBERT.

#### F Per language classifications

This appendix provides extended confusion matrix analyses complementing the summary in Section 6.2. We include both zoomed-in views of key misclassifications and complete confusion matrices for each classifier. Each matrix presents the language classification results for each ground truth language on all texts in the test set by reporting average values over three random seeds, with a maximum of 250 predictions per language.



(a) mBERT + MLP classifier.



(b) mBERT + CNN classifier.

Figure 6: Zoomed-in confusion matrices highlighting the main misclassifications for two classifier architectures.

## F.1 Zoomed in versions of the Confusion Matrices

Figures 6a and 6b show zoomed-in sections of the confusion matrices for the MLP and CNN classifiers (the Linear version is in Figure 2 of the main text).

Across all classifiers, the strongest confusion occurs among the South Slavic languages: Bosnian, Croatian, and Serbo-Croatian. This tendency is linguistically grounded: extensive empirical evi-

dence demonstrates that these varieties form a single pluricentric language, sharing an almost identical grammatical and morphological system (Bailyn, 2010; Bunčić, 2021). Interestingly, Serbian exhibits fewer such errors (Figure 6a) despite its linguistic proximity, suggesting that orthographic differences or dataset imbalance<sup>4</sup> may contribute to clearer separability.

As mentioned in the main text, the less frequent misclassifications arise between languages with strong lexical or geographical ties, such as German–Bavarian (Rowley, 2011), French–Occitan (Klingebiel, 2017), and Malay–Indonesian<sup>5</sup>. These cases likely reflect genuine linguistic overlap rather than model failure. In contrast, a recurring but minor pattern (less than 15 instances across all heads) involves Yoruba texts being labelled as English; however, this likely attributed to mislabeled samples containing predominantly English content like the one seen in Appendix A.

Languages that are typologically or geographically distant (such as Yoruba, Malay, and German) show almost no confusion with others, highlighting the effectiveness of the shared mBERT encoder in maintaining cross-lingual feature separation.

While these overall patterns hold across architectures, closer inspection of the individual models reveals distinct tendencies in how errors are distributed, which we detail in the following subsection.

The **MLP model** (Figure 6a) demonstrates the highest degree of confusion within the South Slavic cluster, most prominently mislabel *Croatian* and *Serbo-Croatian* as *Bosnian* (with counts exceeding 170). It also shows the widest spread of misclassifications overall, incorrectly predicting 71 languages (for a total of 161 misclassified instances; see Figure 8). In addition to these dominant errors, smaller overlaps are observed between Indonesian–Malay and Bavarian–German (approximately a dozen cases each), suggesting that while the MLP captures broad nonlinear relationships, it struggles to distinguish between highly similar linguistic varieties.

<sup>4</sup>One possible explanation relates to the pre-training data distribution of mBERT, as Serbian has approximately twice the number of Wikipedia articles compared to the other South Slavic languages (see Table 3).

<sup>5</sup>According to (Sneddon, 2004), Bahasa Indonesia and Bahasa Melayu are standardized national varieties of the same Malay language. While they are largely mutually intelligible, they have differences in vocabulary, grammar, and spelling due to distinct colonial and cultural influences that accompanied their separate standardizations.

The **CNN classifier** (Figure 6b) displays a pattern broadly similar to that of the MLP, with persistent but less severe confusion among Bosnian, Croatian, and Serbo-Croatian (typically 50–90 instances). Unlike the MLP and Linear models, which primarily misassign the label *Bosnian*, the CNN tends to predict *Serbo-Croatian* in these ambiguous cases. This classifier exhibits the strongest diagonal dominance (always exceeding 125 correctly classified instances per language), indicating more robust separation across languages. Overall, it yields 164 misclassifications spanning 66 languages (Figure 9), underscoring its comparatively better generalization.

The **Linear model** (Figure 2) most frequently predicts *Bosnian* as *Croatian* or *Serbo-Croatian* (over 100 instances each). Despite this, it effectively distinguishes typologically distant languages such as English, French, and Yoruba. Moreover, it performs comparatively well on certain closely related pairs where it produces fewer than ten misclassifications: such as, *Malay–Indonesian* and *Bavarian–German*. In total, the Linear classifier accounts for 136 misclassifications across 68 languages (Figure 7), suggesting that, while limited in representational capacity, it maintains a degree of robustness in separating languages with distinctive orthographic or lexical characteristics.

## F.2 The Complete Confusion Matrices

Figures 7–9 display the full per-language confusion matrices for each classifier. These show similar overall trends across architectures: strong diagonal dominance and limited off-diagonal confusion. The MLP produces the largest number of distinct misclassifications (71 languages affected), the CNN achieves the most robust separation, and the Linear model performs comparably despite its simpler architecture.

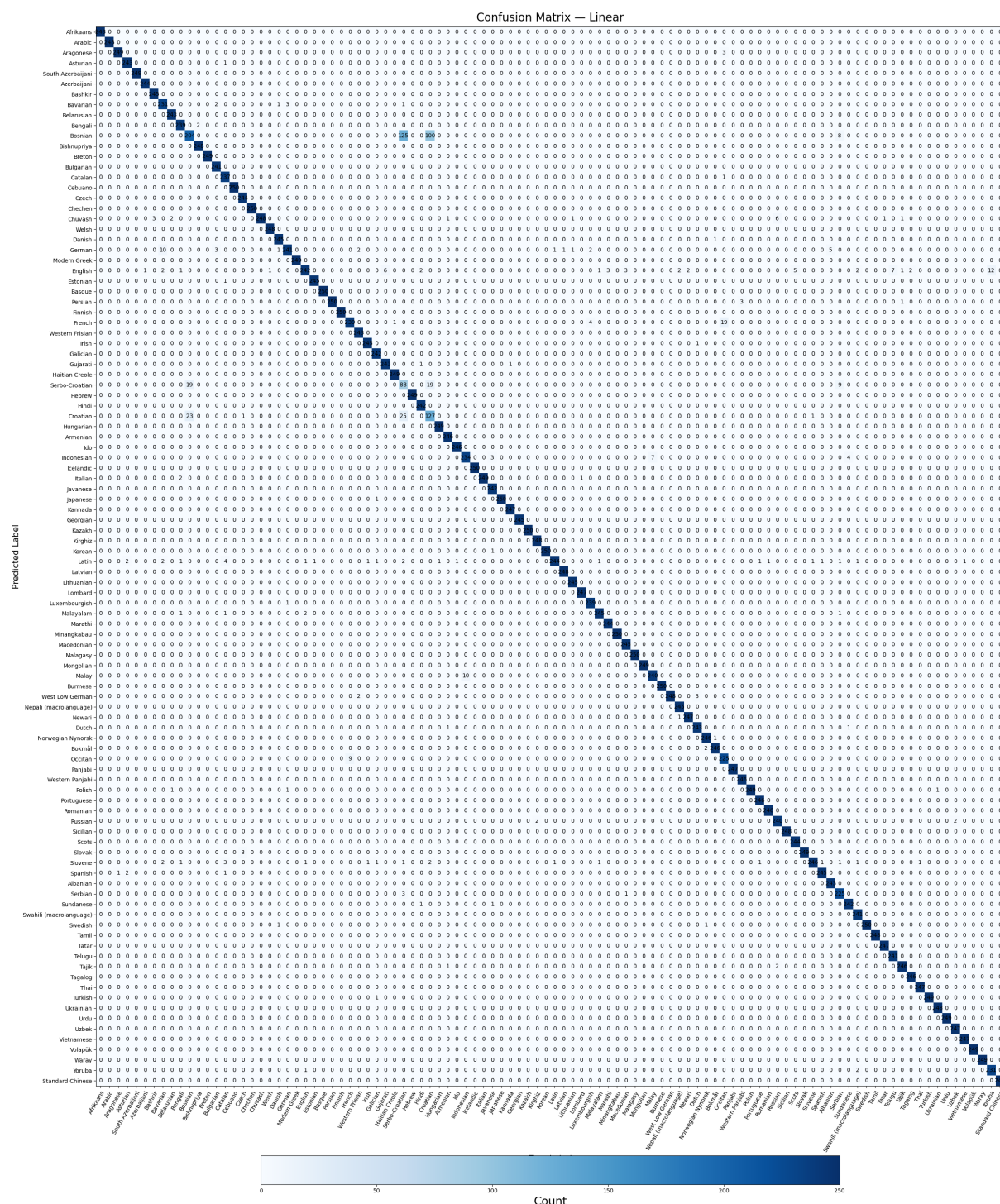


Figure 7: Language classification results of the mBERT + Linear model configuration per language. Each value is the average taken over all three seeds.



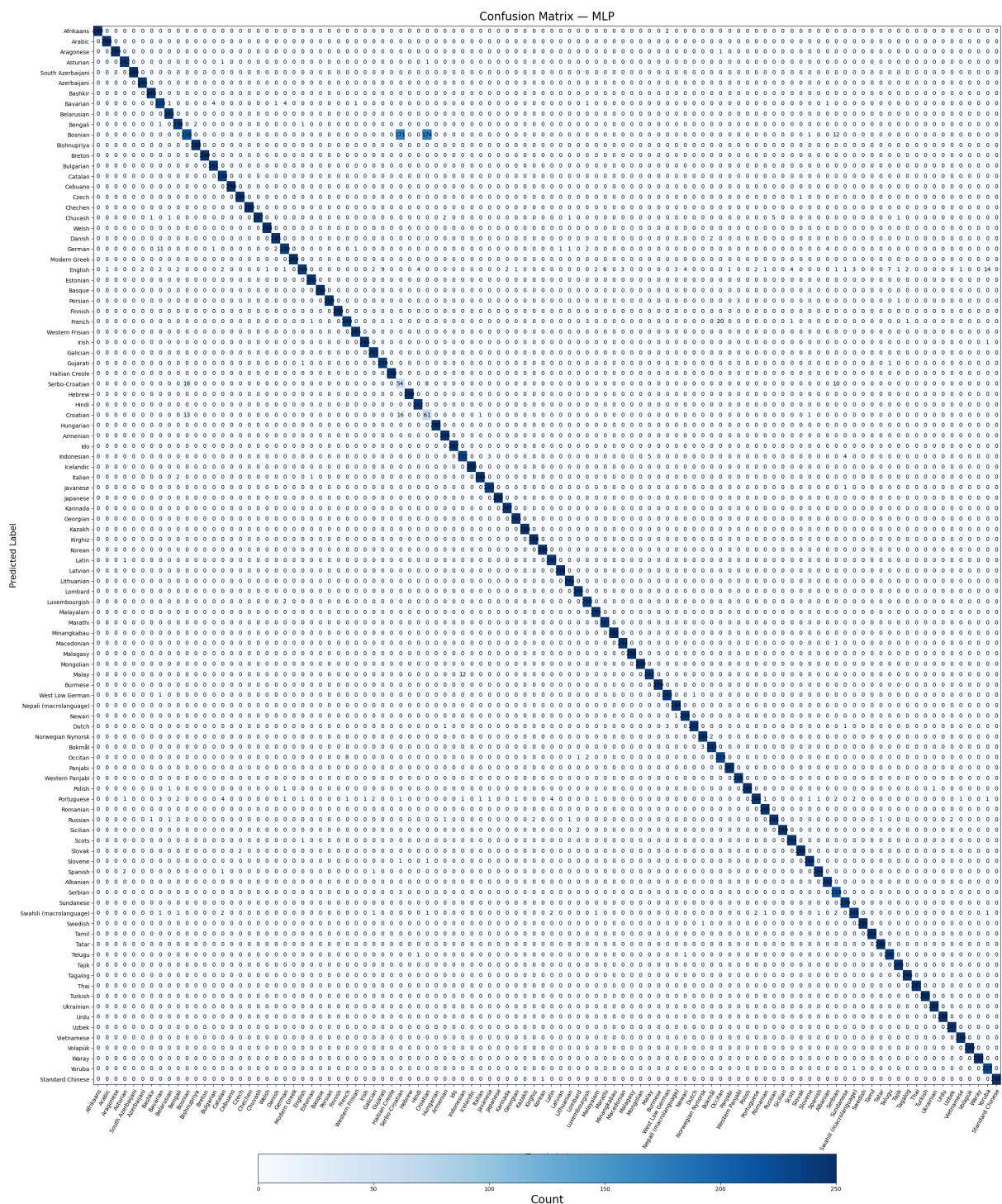


Figure 8: Language classification results of the mBERT + MLP model configuration per language. Each value is the average taken over all three seeds.

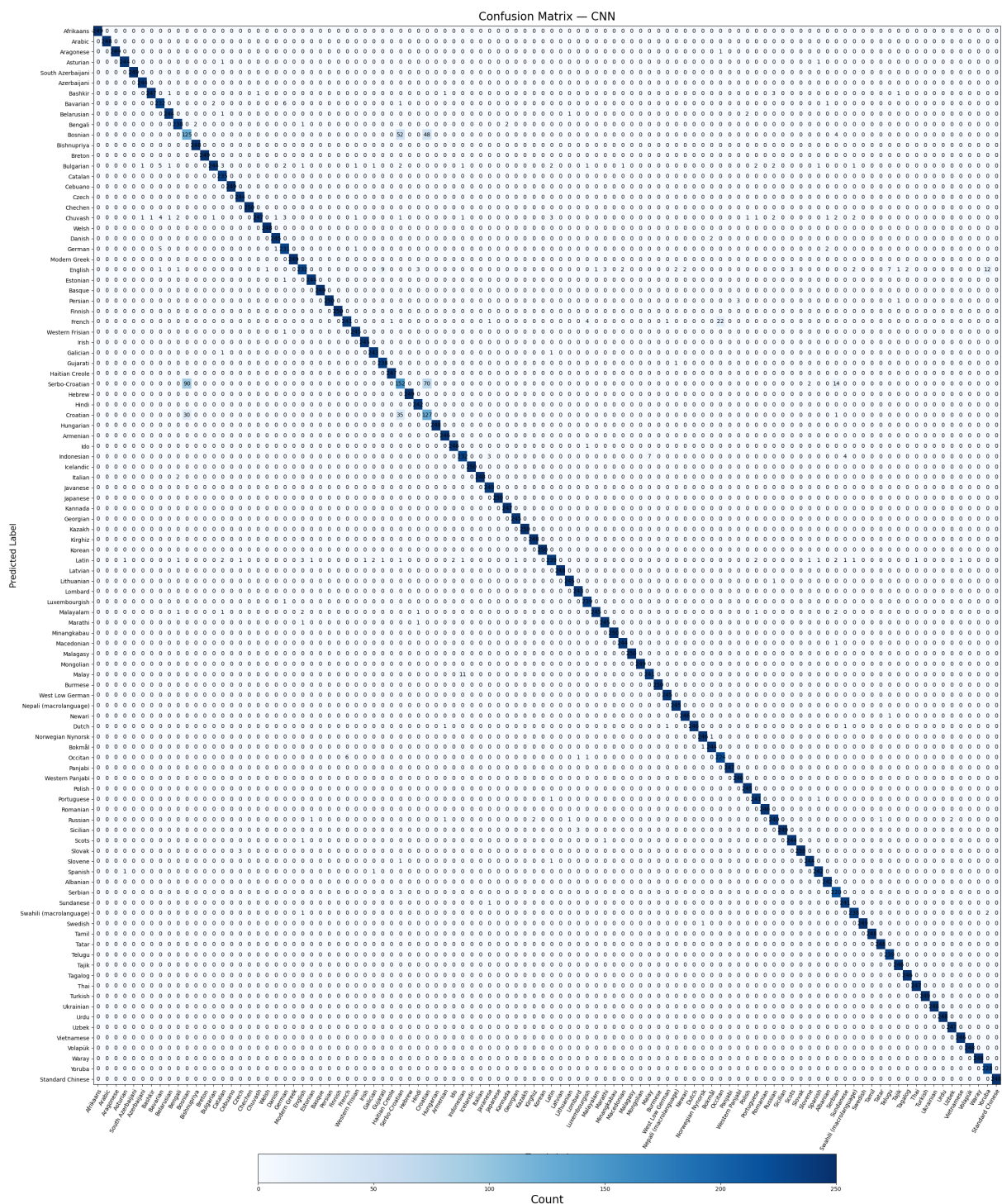


Figure 9: Language classification results of the mBERT + CNN model configuration per language. Each value is the average taken over all three seeds.